NUS HIGH SCHOOL of Math & Science

National Mathematical Olympiad of Singapore

Date: 16 JULY 2024

Time Given: 1 hour 30 minutes

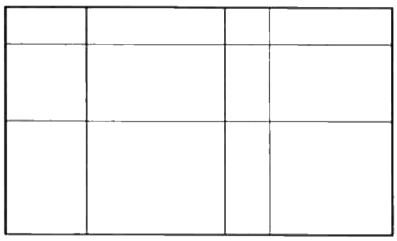
Instruction to Candidates

1. Do not open the booklet until you are told to do so.
2. Answer ALL 35 questions.
3. Write your answers in the answer sheet provided and shade the appropriate bubbles below your answers.
4. No steps are needed to justify your answers.
5. Figures are not necessarily drawn to scale.
6. Questions 1 – 10 are worth 1 mark each.
7. Questions 11 – 20 are worth 2 marks each.
8. Questions 21 – 30 are worth 3 marks each.
9. Questions 31 – 35 are worth 4 marks each.
10. No marks will be deducted for wrong answers.
11. Unanswered questions will not get any marks.
12. No calculators or mathematical instruments are allowed.

Questions 1 to 10 are worth 1 mark each.

1. Evaluate $(19\times19-12\times12)\div\left(\frac{19}{12}-\frac{12}{19}\right)$.

2. Find the number of rectangles (including squares) in the following diagram.



3. There are altogether 30 coins of 20 cents and 50 cents with a total worth of \$9.90. How many 20-cent coins are there?

4. Given that a, b, c and d are different prime numbers, and $a+b+c=d$. Find the smallest possible value of $a\times b\times c\times d$.

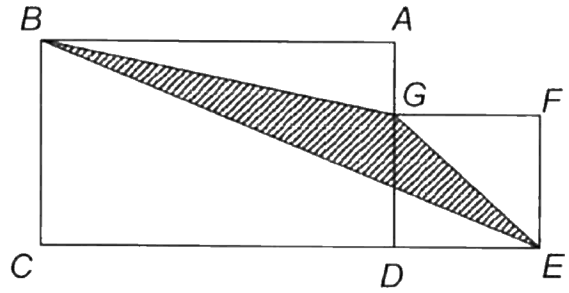
5. Town A and Town B are connected by a straight road. At 9am, car X travels from Town A to Town B at a constant speed, and car Y travels from Town B to Town A at a constant speed. They meet at the place 30 meters away from the midpoint of the two towns on this road. Given that the speed of car X is $\frac{5}{6}$ of the speed of car Y. Find the distance between Town A and Town B in meters.

6. $\overline{ABC}$ and $\overline{DEF}$ represent two 3-digit numbers where A, B, C, D, E and F are distinct digits, and A and D are non-zero. In the following sum, what is the smallest $\overline{ABC}$?

$$\begin{array}{r} A\ B\ C \\ +\ D\ E\ F \\ \hline 5\ 6\ 7 \end{array}$$

7. A tin of cookies was shared among Alan, Barbara and Connie in the ratio of $4:6:14$. If the cookies were shared equally among them instead, Barbara would have 8 more pieces of cookies. Determine the total number of cookies in the tin.

8. The diagram below shows a rectangle *ABCD* and square *DEFG* side by side. The area of the rectangle is 150 cm^2 with the ratio of length to width being 3 : 2. The area of the square is 64 cm^2. Find the area of the shaded region in cm^2.



9. Mark and James can assemble 80 toy trains together in two hours. Mark alone can assemble 32 toy trains in 80 minutes. Assuming their rates of working remain constant throughout, how many minutes does James alone take to assemble a total of 64 toy trains?

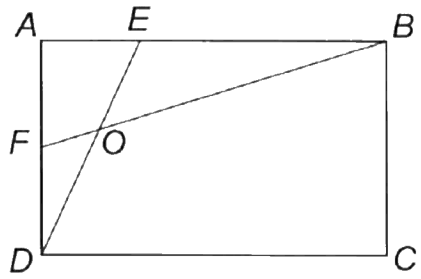
10. In a class of 40 students, the average score for a Mathematics test is 76. Two of the students, Alicia and Betty, leave the class. The average score of the remaining students increases to 77. Given that Alicia scores 2 more marks than Betty, what is Alicia's score?

Questions 11 to 20 are worth 2 marks each

11. Sammy placed an online order for 275 plates and some cups for her new café. When her order was delivered, she inspected and found that 8% of her plates and 4% of the cups were chipped. The other items, made up of 94.9% of the total number of items ordered, were delivered in good condition. How many cups did Sammy buy?

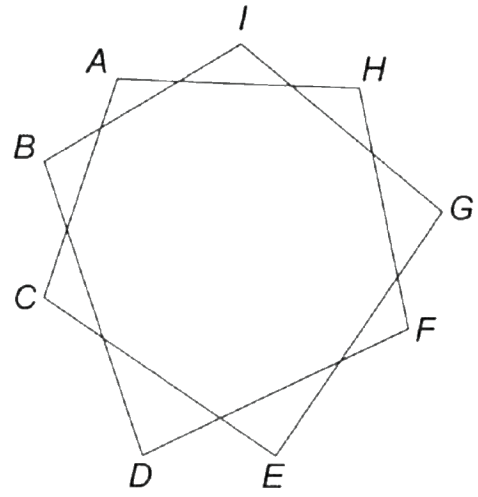
12. A palindrome is a number that reads the same backwards and forwards, such as 1001 and 54345. Find the smallest 2-digit number $\overline{AB}$ such that the sum $\overline{AB}+\overline{BA}$ is not a palindrome.

13. In the diagram below, area of rectangle $ABCD$ is 60 cm^2. Given that $BE = 2AE$ and $AF = FD$, find the area of quadrilateral $AEOF$ in cm^2.



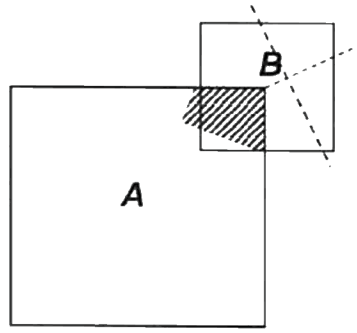
14. At a running race, competitors A, B and C are running at constant speeds. A is 100 meters ahead of B and B in turn is 300 meters ahead of C. B is confident that he can catch up with A in 5 mins while C is certain that he can catch up with B in 10 mins. How many minutes does C take to catch up with A?

15. In the diagram below, find
$\angle A+\angle B+\angle C+\angle D+\angle E+\angle F+\angle G+\angle H+\angle I$ in degrees.

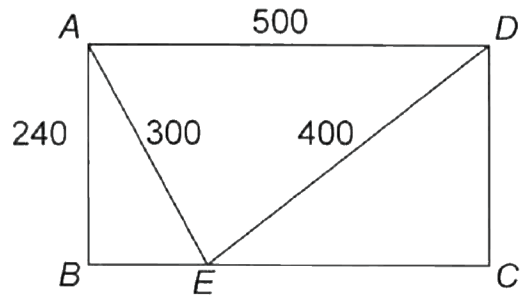


16. It is known that the sum and the difference of two whole numbers are in the ratio 29 : 9. Furthermore, the product of these two whole numbers is 1710. Find the smaller number among these two numbers.

17. In the following diagram, squares A and B are of different sizes. The intersection point of diagonals in B coincides with a vertex point of A. It is known that the overlapping area (shown as the shaded region) is $\frac{1}{9}$ of the area of square A. If the length of a side of square A and the length of a side of square B are in the ratio of $m:n$, where $\frac{m}{n}$ is in its simplest form, find the value of $m+n$.



18. In the diagram below, $ABCD$ is a rectangle with length of 500 meters and width of 240 meters. E is a point on BC such that $AE = 300$ meters and $DE = 400$ meters. Peter travels along the rectangle via the route $A \to B \to C \to D \to A$, and then rest for 20 seconds at point A. Quintus also travels along the triangle via the route $A \to E \to D \to A$ without resting. If both travel at the speed of 1 meter per second and start travelling from A at the same time. Find the time taken (in mins) that they will next meet again at A.

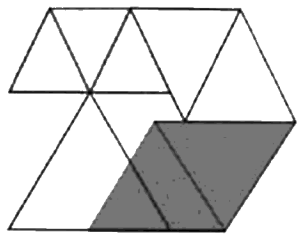


19. In the calendar of a certain month, there are 5 Mondays, 5 Tuesdays and 5 Wednesdays. On which day is the last day of the month?
[If your answer is Monday, then shade your answer as "1";
If your answer is Tuesday, then shade your answer as "2";
If your answer is Wednesday, then shade your answer as "3";
If your answer is Thursday, then shade your answer as "4";
If your answer is Friday, then shade your answer as "5";
If your answer is Saturday, then shade your answer as "6";
If your answer is Sunday, then shade your answer as "7".]

20. In a certain school, 172 students passed English, 179 students passed Mathematics, and 101 students passed Science but failed Mathematics. Given that all 378 students in the school passed at least one of the three abovementioned subjects, how many students passed English only?

Questions 21 to 30 are worth 3 marks each

21. The 6-sided figure below is composed of 9 equilateral triangles. Given that the side of the smallest equilateral triangle in the middle is 1cm, find the perimeter of the 6-sided figure in cm.



22. There are 11 numbers in a certain sequence, as shown below:

$$a, b, c, d, e, f, g, h, i, j, k.$$

The sum of these 11 numbers is 2024. If the difference between any two consecutive terms is a constant, i.e.

$$b-a=c-b=d-c=e-d=f-e=g-f=h-g=i-h=j-i=k-j,$$

what is the value of $a+k$?

23. From a solid cube with each side being 6 cm long, Figure 2, which is composed of 7 identical cubes, is removed so that each face has a 2 cm$\times$2 cm square hole. The remaining figure is Figure 1. Find the total surface area of Figure 1 in cm^2.

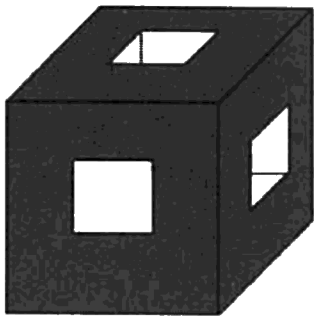


Figure 1

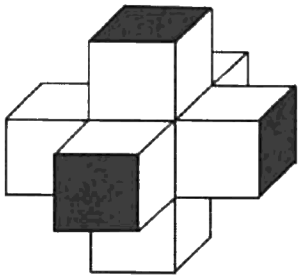


Figure 2

24. Let us generate a sequence of numbers by multiplying 2 to the preceding term to get the next term, with 2 as the first term, as follows:

$$2, 4, 8, 16, 32, 64, 128, 256, \ldots$$

What is the remainder when the sum of the 40th term and the 67th term of this sequence is divided by 5?

25. In a certain population,

- The ratio of the number of women who wear glasses to the number of women who do not wear glasses is $5:8$,
- The ratio of the number of men who wear glasses to the number of men who do not wear glasses is $3:4$,
- The ratio of the number of people who wear glasses to the number of people who do not wear glasses is $7:10$.

If the ratio of the number of women who wear glasses to the number of men who wear glasses is $m:n$, where $\frac{m}{n}$ is in its simplest form, find the value of $m+n$.

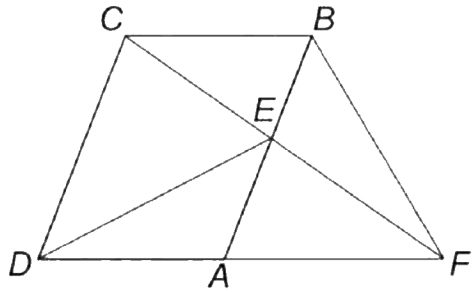
26. A furniture retailer had a total of 7270 dining table and sofa sets for sale at the start of the month. He sold $\frac{3}{5}$ of his dining tables and 72% of his sofa sets. He imported an additional 52 dining tables. In the end, the number of sofa sets he had left was $16\frac{1}{3}\%$ of the number of dining tables left. How many sofa sets did he have at first?

27. A particular brand of detergent is sold in three different sizes at a supermarket as shown in the table below:

	Amount of detergent per packet	Cost per packet
Small	2 litres	\$2.50
Medium	5 litres	\$6
Large	9 litres	\$10

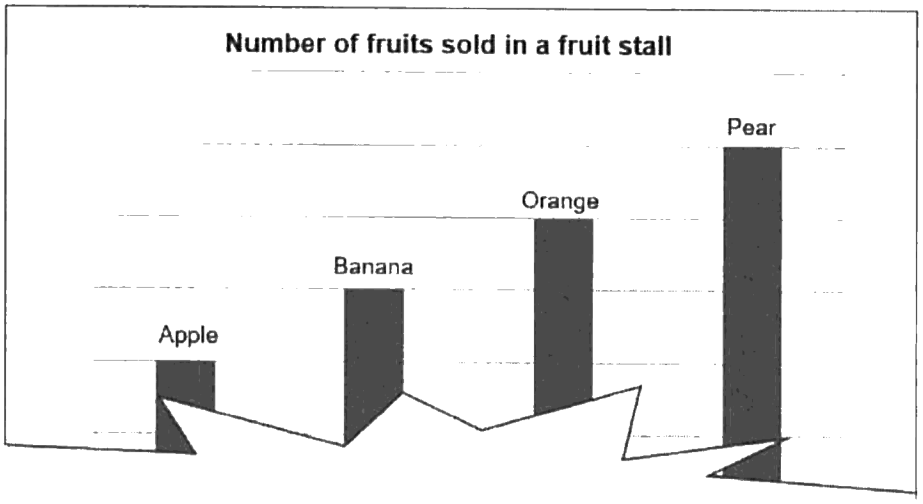
What is the maximum number of litres of the detergent that can be purchased with \$52?

28. In the diagram below, DAF is a straight line and $ABCD$ is a parallelogram with CF and AB intersecting at E. If the area of ΔADE is 10 cm^2, find the area of ΔBEF in cm^2.



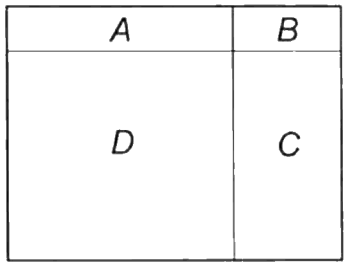
29. There are 20 straight lines drawn on a plane, among which 3 lines are parallel to each other. What is the maximum number of regions that can be divided by these 20 lines?

30. A fruit seller sells only apples, bananas, oranges and pears. The bar chart below shows the number of fruits sold in his fruit stall in the month of January. However, the bottom part of the bar chart is torn off. If pears made up 30% of his January sales of fruits and x% of the fruits sold are apples, find the value of x.



Questions 31 to 35 are worth 4 marks each

31. In the rectangle below, figures A, B and C are rectangles and figure D is a square. If the sum of areas of A and B is 23 cm^2 and the sum of areas of B and C is 44 cm^2, given that all length of rectangles are positive integers, find the maximum area of the square D in cm^2.



32. Let us consider the product of two numbers, where x and y are positive whole numbers:

$$\left(1+\frac{2}{x}\right)\left(1+\frac{2}{y}\right).$$

In some cases, the product above produces a number of the form $1+\frac{1}{z}$ for
some positive whole number z, such as the following:

$$\left(1+\frac{2}{8}\right)\left(1+\frac{2}{30}\right)=1+\frac{1}{3},$$

$$\left(1+\frac{2}{18}\right)\left(1+\frac{2}{40}\right)=1+\frac{1}{6}$$

Find the smallest product xy such that $\left(1+\frac{2}{x}\right)\left(1+\frac{2}{y}\right)=1+\frac{1}{5}$.

33. Malik purchased a total of 12 fruits, consisting of both apples and oranges, for \$14.70. One apple costs \$0.03 more than one orange and Malik bought more apples than oranges. How much, in cents, did Malik pay for the apples alone?
Note: The smallest denomination is 1 cent. There are no fractions of cents.

34. A number is formed by writing a "1" after every n "0"s, with the number of "0"s written increased by one more after writing each "1" (where $n = 0, 1, 2, 3, \ldots$). The first 36 digits of this number is shown below:

$$110100100010000100000100000010000000\ldots$$

Determine the number of "0"s in the first 2024 digits.

35. A meteorological report shows the following records of a certain number of days.

- It rained 12 times, either in the morning, afternoon or evening. (For clarity, if it rained the entire day, this is counted as having rained 3 times in total).
- There were 12 dry mornings.
- There were 13 dry afternoons.

What is the minimum possible number of days the meteorologist could have recorded?

- THE END -